

NGUYEN LE GIA HUY

Frontend Developer Intern / Software Interface Engineer

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PROFESSIONAL SUMMARY

Software Engineering graduate specializing in **Modern Frontend Architecture, Web Performance Optimization & User Experience (UI/UX)**. Hands-on experience developing responsive web and mobile applications with **Next.js App Router, React, Flutter, Tailwind CSS, and TypeScript Strict Mode**. Proven expertise in client-side Edge AI with **TensorFlow.js (WebGL acceleration)**, Core Web Vitals optimization (CLS = 0, Lighthouse 95+), scientific LaTeX formula rendering with **MathJax**, zero-asset audio synthesis with **Native Web Audio API**, reactive state management (**Zustand, Riverpod**), and client security (**JWT, OAuth 2.0, RBAC, XSS Sanitization**). **Available for full-time internship and employment.**

FRONTEND SKILLS

Languages:	TypeScript, JavaScript, Dart, HTML5/CSS3, Python, SQL
Frontend & Mobile:	Next.js App Router, React, Flutter, Tailwind CSS, SCSS, Framer Motion, PWA
State & Client AI:	Zustand, Riverpod, TanStack Query, TensorFlow.js (WebGL), kuromoji.js, MathJax, Web Audio API
Quality & Tooling:	Playwright E2E, Jest, Figma, WCAG 2.1 AA, Lighthouse, Git, Vercel, REST/Socket.IO

FEATURED PROJECTS

03/2026 – Present

HyperGlot — AI Japanese Learning Ecosystem & Monorepo (Next.js & Flutter)

Role: **Lead Frontend / Mobile Architect** | Monorepo: github.com/nlgiahuy01112003/HyperGlot

Next.js · Flutter · Riverpod · TanStack Table · Recharts · TensorFlow.js (WebGL) · Playwright · WCAG 2.1 AA

Enterprise-grade multi-platform Japanese learning platform featuring a Next.js learner portal, Admin Dashboard, Flutter Mobile App, and Chrome Extension under a Zero-Cost API strategy.

- **Edge AI & Client-Side NLP (Zero-Cost API):** Deployed local kanji stroke recognition using TensorFlow.js (WebGL hardware acceleration) and kuromoji.js tokenization directly inside the browser, eliminating 100% of network latency and cloud API costs.
- **Turborepo Monorepo Multi-Platform UI:** Engineered `apps/web` (Next.js), `apps/admin` (TanStack Table & Recharts analytics), and Manifest V3 Chrome Extension sharing design tokens and strict type contracts.
- **Mobile Offline-First & SRS Review:** Built a Flutter mobile application with Riverpod StateNotifier and local Isar DB, delivering silky-smooth 60 FPS offline flashcard reviews.
- **Accessibility & Cross-Browser E2E Testing:** Complied with international WCAG 2.1 AA accessibility standards (keyboard/screen reader support) and built automated Playwright E2E suites covering learner journeys.

09/2025 – 03/2026

TechStore (Vanfio) — Next.js E-Commerce Storefront & UX Optimization (React)

Role: **Frontend / UI Engineer** | Demo: website-for-aunt.vercel.app | Source: github.com/nlgiahuy01112003/techstore

Next.js · React · TypeScript Strict · Zustand · SCSS · Framer Motion · Spotlight Search · Lighthouse 95+

Production electronics e-commerce storefront spanning 27 responsive routes, reactive state orchestration, and fluid 60 FPS interactions.

- **Core Web Vitals & Render Tuning:** Eliminated client hydration waterfalls and Cumulative Layout Shift (CLS = 0), earning **95+ Google Lighthouse scores** across low-tier mobile devices.
- **Reactive State Architecture (Zustand):** Designed modular Zustand stores (auth, cart, compare, recently viewed, live chat) with localStorage persistence and an interactive 60 FPS cart drawer.
- **Spotlight Search & Keyboard UX:** Built keyboard-accessible Spotlight Search (Ctrl/Cmd+K) with a 350ms debounce, SKU indexing, VI/EN/ZH synonym suggestions, and sub-50ms query resolution.
- **Client Defense & RBAC Route Guards:** Enforced client-side security with rotating JWTs, RBAC Route Guards across 27 routes, strict XSS sanitization, and Workbox PWA offline asset caching.
- **Frictionless Checkout & MoMo UX:** Engineered a fluid checkout workflow with duplicate click locking, automated 10-minute voucher reservation countdowns, and real-time MoMo IPN visual indicators.

OTHER CORE PROJECTS

08/2026 – 09/2026

AI Mock Exam Platform — SFIA L3-4 AI Assessment Engine (Next.js, Pyodide & LaTeX)

Role: **Frontend / UI Developer** | Demo: ai-exam-live | Source: GitHub

Next.js | Pyodide WebAssembly | Tailwind CSS | MathJax (LaTeX) | Native Web Audio API | Gemini AI Mentor

- **In-Browser Python Sandbox & MathJax:** Integrated **Pyodide WebAssembly** to run candidate Python/NumPy code directly in the browser alongside **MathJax** LaTeX rendering across **1,020+ questions** without frame drops.
- **Zero-Asset Audio Synthesis (Web Audio API):** Synthesized interactive UI sound effects (click, correct chime, timer alarm) via Native Web Audio API, **saving 100% of external audio asset payload**.
- **Pro Exam Room Experience:** Crafted a distraction-free UI with Dark/Light theme toggle, hotkey navigation (1-4 answers, F flag question, Enter next), visual progress indicators, and an AI Smart Mentor assistant.

03/2024 – Present

POMI Milk Tea — Real-Time F&B Management Interface (React & Maps)

Role: **Frontend Developer** | Source: github.com/nlgiahuy01112003/pomi_web

React | Sass | TanStack Query | Socket.IO | Leaflet / Goong.io | Flutter | Cloudinary

- **Sub-Second Real-Time DOM Updates:** Engineered a kitchen display dashboard binding incoming Socket.IO payloads directly into React state under 800ms without costly full DOM tree reconciliations.
- **Interactive Geospatial Mapping:** Integrated Leaflet and Goong.io API to visualize store branches and automatically compute delivery radius based on real-time customer GPS coordinates.
- **Low-End POS Tablet Optimization:** Applied route-level code splitting and Cloudinary image optimization, ensuring instant POS touch interactions on budget tablets.

06/2026 – 07/2026

OptiBot RAG System (KB-Harvest) — Grounded AI Assistant CLI & RAG Knowledge Pipeline

Role: **Python / AI Engineer** | Source: github.com/nlgiahuy01112003/kb-harvest

Python | Google Gemini API | BeautifulSoup4 | Markdownify | Docker | GitHub Actions

- **Token-Efficient Content Processing:** Pipeline scrapes Zendesk Help Center documentation, transforming raw HTML into clean Markdown structure (saving > 40% LLM context tokens).
- **Grounded Knowledge Retrieval:** Configured Gemini File Search RAG to deliver accurate technical answers via terminal with source URL citations; automated via **GitHub Actions cron (02:00 UTC)**.

EDUCATION

09/2021 – 03/2026

Engineer's Degree in Information Technology (Software Engineering)

HUTECH University — GPA: 2.86/4.0

- **Core foundation:** Data structures & algorithms, object-oriented programming (OOP), relational databases, operating systems, computer networks, software architecture, and automated testing.
- **Frontend practice:** Developed responsive web and mobile interfaces with emphasis on interaction flows, API integration, UI testing, and Git collaboration.

CERTIFICATIONS & LANGUAGES

- **CEFR B1 English — HUTECH:** General English Test at Intermediate Level; issued Dec 4, 2025.
- **Udemy — Flutter Development Bootcamp with Dart:** Dr. Angela Yu; Dart OOP, Flutter State Management & APIs.
- **Communication & Teamwork — HUTECH:** Issued Apr 16, 2025; technical problem solving and cross-team collaboration.
- **Udemy — Full-Stack Web Development Bootcamp:** Dr. Angela Yu; React, Node.js, Express, REST APIs.
- **Udemy — 100 Days of Code™ Python Pro Bootcamp:** Dr. Angela Yu; Python OOP, Web Scraping, Automation.

PROFESSIONAL EXPERIENCE

12/2025 – 06/2026

IT Helpdesk / Manual Tester & System Deployment

EMPOWER SC

- Supported stability of the **Dimchan All-in-One** POS/ERP ecosystem across F&B/retail businesses; maintained over 95% on-site resolution.
- Executed end-to-end manual testing on web and mobile releases; documented structured defect tickets with reproduction logs to accelerate bug triage.

08/2025 – 10/2025

IT Support & System Deployment Specialist (Project Contract)

HCMC Digital Transformation Center (DXCENTER)

- **Business Process Training (First 2 Weeks):** Recruited for web technical expertise under a municipal digital project; mastered end-to-end workflows of 2 public portals (motcua.tphcm.gov.vn, qlvb.tphcm.gov.vn).
- **Field Stationed at Binh Khanh Ward (1.5 Months):** Directly trained civil servants and citizens on both portals; handled user inquiries, reproduced edge-case bugs, and dispatched structured bug reports to L2 Developers.
- **Hardware, Networking & Digital Signatures:** Troubleshot PC hardware, office LAN networks, and configured digital authentication/signatures; fulfilled deployment contract milestones.